

Pairs of straight thorns keep animals away

TANGLED IN A TRAP
Thorns on flexible stems snag passing animals and give them a painful lesson, teaching them not to get too close again. Thorns can be either straight, as on this rose, or they can be curved, which makes them dig in when they are pulled. Some plants have thorns pointing in both directions along their stems. If an animal gets tangled in the plant, the thorns catch no matter which way it pulls.

Thistle in flower

Unopened flower head is protected by spiny bracts

Woolly thistle

ARMORED FLOWERS

Thistles are extremely successful plants, partly because they have an effective seed dispersal system (p. 27), and partly because they are very well defended. Most thistles have spines on their stems as well as on their leaves. They also have spiny flaps, known as bracts, that protect their developing flower heads. The spines keep most animals away, although they do not deter sap-sucking insects such as aphids, which often feed in large numbers on thistles.

DEATH BY DROWNING

Pairs of leaves on the stem of the teasel meet to form little cups that fill with water after rain. Each cup acts like a moat to protect the plant. Snails and insects try to climb the plant to feed on its young leaves but, faced with the water, they either turn back or fall into the moat and drown.

Spiny bracts protect the developing flower head

Young, spiny flower head

Teasel

Holly leaves

Insects drowned in moat formed at the point where the paired leaves meet

Teasel leaves, seen from above

Spines protect the stem

EXTRA WEAPONRY

Holly leaves are not only very leathery, but they also have tough spines all around their edges. Compared to the leaves of deciduous trees, which lose their leaves every fall, they suffer very little from animal attack. These leaves are from a variety of holly that has been cultivated to produce extra spines. On holly trees, the lowest leaves are usually the spiniest. Those nearest the top may have no spines at all, since they are in less danger of being eaten.